

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- AERMINUTE
- Expectation
 - Understand the role of AERMINUTE and be able to use it
- Readings: AERMINUTE User guides

AERMINUTE

- Average 1-min ASOS data to 1 hour for AERMET use
- Detailed explanation of TD-6405 ASOS data
 - AERMINUTE user guide page 4
- ASOS data download
 - <ftp://ftp.ncdc.noaa.gov/pub/data/asos-onemin/>
 - Look for folders start with 6405
 - File format: 6405 + 0 + station call sign + YYYY + MM
 - To find a station near your study area:
 - https://www.faa.gov/air_traffic/weather/asos/
 - Look for only ASOS stations

AERMINUTE

- AERMINUTE uses an input file (not interactive)
- The input file requires at least four sections
 - STARTEND: start and end month of data processing
 - IFWGROUP: if/when this station measures ice free wind (IFW)
 - Ice affect cup and vane sensors, not heated sonic anemometers
 - Installation date of IFW wind instrument:
 - https://www.weather.gov/media/asos/ASOS%20Implementation/IFW_stat.pdf
 - DATAFILE: list of input data files
 - Starts with “DATAFILE STARTING”, end with “DATAFILE FINISHED”
 - List all 1-min ASOS data files
 - OUTFILES: list of output data files
 - Starts with “OUTFILES STARTING”, end with “OUTFILES FINISHED”
 - HOURFILE: name of output hourly data for AERMET

Sample AERMINUTE input

STARTEND 1 2017 6 2017 → Start from Jan 17 to Jun 17
IFWGROUP Y 5 31 2007 → Yes, measures IFW wind,
starting from 5/31/07
DATAFILE STARTING
64050KMCO201701.dat
64050KMCO201702.dat
64050KMCO201703.dat
64050KMCO201704.dat
64050KMCO201705.dat
64050KMCO201706.dat
DATAFILE FINISHED
OUTFILES STARTING
HOURFILE MCO.dat → Output to “MCO.dat”
OUTFILES FINISHED

A list of all input 1-min ASOS data

Output

- In addition to the specified output, other output data are also available:
 - aerminute.log: a log file
 - good_records.dat: input data passed QA/QC
 - bad_records.dat: input data failed QA/QC
 - check_records.dat: input data seems suspicious (thus not used), but just might be useful

Other options

- AERMINUTE also has several other options available
 - Include 5-min ASOS data
 - Compare averaged data with regular hourly measurement data
 - And output comparison results to a file
 - This is because ASOS data commonly contains substantially more calm record (wind speed < 1 m/s) than regular hourly data, so it's worth to compare the two
 - Provide a summary file for each hour